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Uncovering Credit Usage Patterns Through Customer Clustering:

An Applied K-Means and DBSCAN Analysis

Customer segmentation allows credit card companies to better understand their customers' usage and behavior. Through grouping customers with similar characteristics, companies can design targeted advertisements, improve credit features, and manage their risk factors more strategically. Unsupervised clustering methods such as K-Means and DBSCAN, as executed in this project, are well fit for this task, as they cover structures in customer data without predetermining categories or labels. The goal of our analysis was to inform marketing strategies and product offerings for the bank's existing customer base.

Our framework identified four unique customer segments including **Pay-in-Full** and **Everyday Users**, who transact frequently with low or moderate balances, and may be best targeted through rewards programs or premium cards with higher annual fees. **Stressed Borrowers**, characterized by high utilization and high revolving balances, may be better served through balance transfer offers, structured repayment plans, or interest-based products. Finally, **Intermediate Revolvers,** who carry a meaningful balance but utilize less credit than Stressed Borrowers and may benefit from sustained credit line increase management.

Dataset Overview & Variable Types

The dataset provided has 10,127 records with 21 columns, representing customer demographics, account duration, credit usage, and purchase behavior. We first inspected the data to confirm that there were no missing or duplicate values that could negatively affect our clustering results later on.

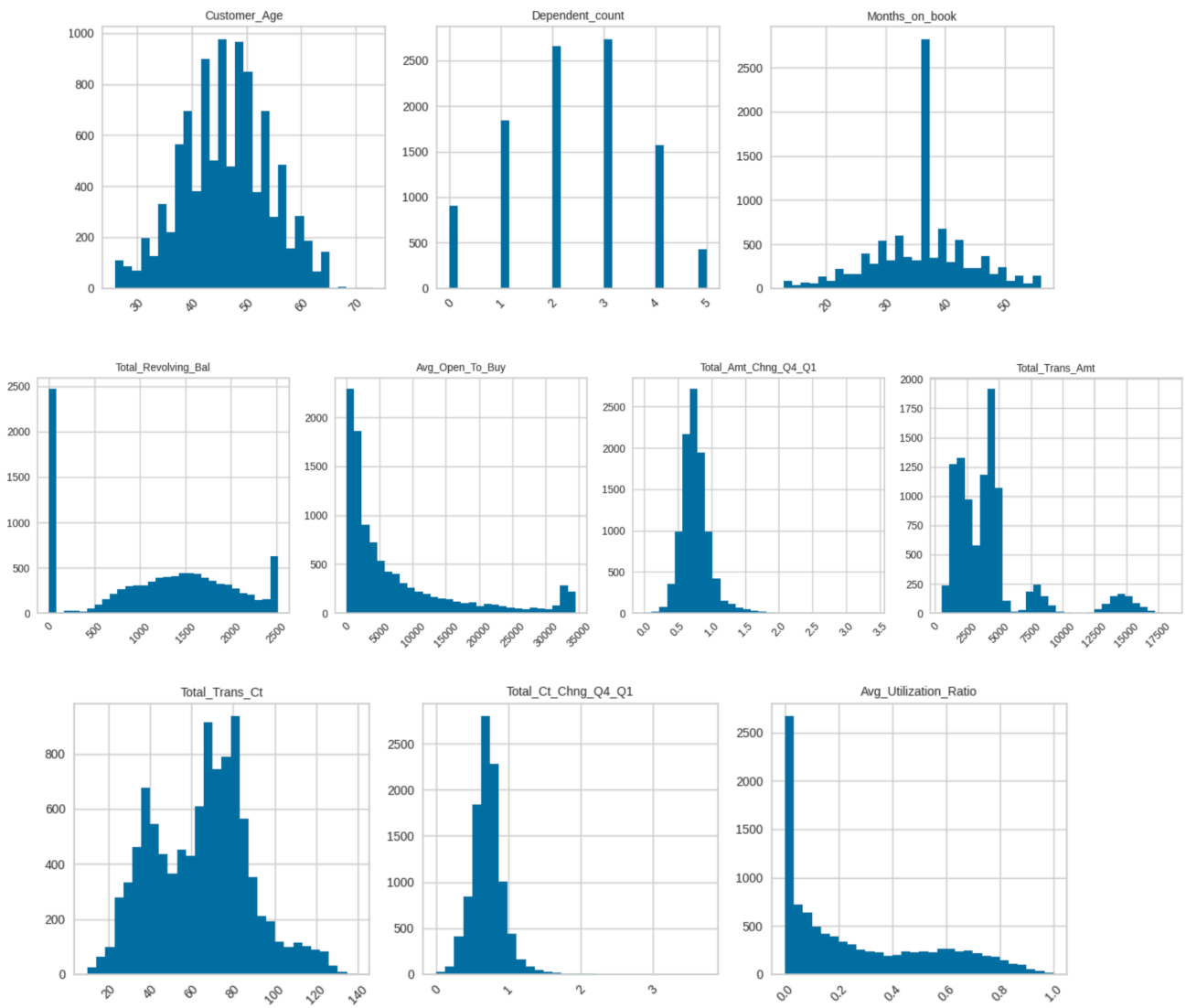
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10127 entries, 0 to 10126
Data columns (total 21 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   CLIENTNUM                 10127 non-null  int64
 1   Attrition_Flag            10127 non-null  object
 2   Customer_Age              10127 non-null  int64
 3   Gender                    10127 non-null  object
 4   Dependent_count           10127 non-null  int64
 5   Education_Level           10127 non-null  object
 6   Marital_Status            10127 non-null  object
 7   Income_Category           10127 non-null  object
 8   Card_Category             10127 non-null  object
 9   Months_on_book            10127 non-null  int64
 10  Total_Relationship_Count  10127 non-null  int64
 11  Months_Inactive_12_mon    10127 non-null  int64
 12  Contacts_Count_12_mon     10127 non-null  int64
 13  Credit_Limit              10127 non-null  float64
 14  Total_Revolving_Bal       10127 non-null  int64
 15  Avg_Open_To_Buy           10127 non-null  float64
 16  Total_Amt_Chng_Q4_Q1      10127 non-null  float64
 17  Total_Trans_Amt           10127 non-null  int64
 18  Total_Trans_Ct            10127 non-null  int64
 19  Total_Ct_Chng_Q4_Q1       10127 non-null  float64
 20  Avg_Utilization_Ratio     10127 non-null  float64
dtypes: float64(5), int64(10), object(6)
memory usage: 1.6+ MB
None
```

The dataset contains15 numeric variables and 6 categorical variables. Since our intended clustering methods rely on distance calculations, we will focus the analysis on numeric features that directly depict customer patterns. Additionally, we will encode our categorical data (see below in "Categorical Variable Encoding" section).

Before moving further, we removed two variables since they were not suited for clustering. These variables are “CLIENTNUM”, which is a unique ID for each customer, as well as “Attrition_Flag”, which is a customer outcome variable instead of a feature variable.

Exploratory Data Analysis & Correlation Analysis

After we identified variables suitable for analysis, we performed descriptive statistics and visualizations on our numeric variables to understand their relationships. Summary statistics (below) revealed large differences in scale across variables, specifically between money measures and ratio-based factors (like credit line utilization). Several variables had right-skewed distributions, especially those related to spending and credit capacity.



Many count-based variables including (Total_Trans_Ct), inactivity (Month_Inactive_12_mon), and customer age (Customer_Age) were highly correlated, as demonstrated below.

		0
Credit_Limit	Avg_Open_To_Buy	0.995981
Total_Trans_Amt	Total_Trans_Ct	0.807192
Customer_Age	Months_on_book	0.788912
Total_Revolving_Bal	Avg_Utilization_Ratio	0.624022
Avg_Open_To_Buy	Avg_Utilization_Ratio	0.538808
Credit_Limit	Avg_Utilization_Ratio	0.482965
Total_Amt_Chng_Q4_Q1	Total_Ct_Chng_Q4_Q1	0.384189
Total_Relationship_Count	Total_Trans_Amt	0.347229

	Total_Trans_Ct	0.241891
Credit_Limit	Total_Trans_Amt	0.171730
Avg_Open_To_Buy	Total_Trans_Amt	0.165923
Contacts_Count_12_mon	Total_Trans_Ct	0.152213
Customer_Age	Dependent_count	0.122254
Contacts_Count_12_mon	Total_Trans_Amt	0.112774
Total_Trans_Ct	Total_Ct_Chng_Q4_Q1	0.112324

To prevent redundancy and prevent correlation from making our results disproportionate, pairs of correlations between the numeric variables were analyzed. We identified correlated feature pairs, and removed one the two based on its relevance to customer behavior. The list of variables removed is displayed below. This resulted in a final numeric set.

```
Excluded features:
['CLIENTNUM', 'Avg_Open_To_Buy', 'Total_Trans_Amt', 'Customer_Age']
```

Categorical Variable Encoding

As mentioned earlier, we see that some variables, such as gender and marital status, can be classified as nominal (they have no inherent order), while other variables (income category and education level), can be classified as ordinal. In order to meaningfully use these categorical datapoints, we encoded each.

```
nominal_pipeline = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

ordinal_pipeline = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("ordinal", OrdinalEncoder(categories=[education_order, income_order])),
    ("scaler", StandardScaler())
])

numeric_pipeline = Pipeline([
    ("scaler", StandardScaler())
])

# Combine transformers/data types
preprocess = ColumnTransformer(
    transformers=[
        ("nom", nominal_pipeline, nominal_var),
        ("ord", ordinal_pipeline, ordinal_var)
    ],
    remainder="drop"
)
```

Because K-Means was selected as the primary clustering framework, these features were not included in the final clustering analysis. The encoded variables were retained for reference and potential use in alternative modeling approaches.

Feature Scaling for Final Dataset

After removing variables for relevance and redundancy, we need to make sure our features are prepared for distance-based algorithms. To do feature scaling, we used Z-score normalization. After scaling, we verified that all variables had zero mean and unit variance, to make sure that certain variables were not overrepresented in distance calculations. Our final dataset for clustering, called **X_prepared_df**, is shown below. It contains 10,127 rows and 18 columns. With preprocessing complete, we can now proceed to K-Means analysis.

K-Means: Overview

K-Means is a distance-based clustering method that forms groups of data by minimizing within-cluster variation across selected behavioral features[1]. The goal of K-means is to develop intuitive, data-supported segments of credit card customers to guide marketing decisions. After a process of trial and error, we selected **revolving balance**, **utilization ratio**, and **transaction count** for final analysis. Together, these variables capture key dimensions of customer behavior including credit usage magnitude, repayment behavior, and transaction activity.

Before running the clustering analysis, we hypothesized that three to four customer groups would emerge: (1) **a financially constrained revolver group** characterized by high utilization and high revolving balances; (2) a **disengaged group** with low utilization and low transaction activity; (3) **a well-off credit gamer group** with high transaction counts and low/no revolving balances; and (4) a potential intermediate group, such as a **less financially constrained revolver segment** with moderate utilization but relatively high revolving balances. Our results generally supported this hypothesis.

K-Means: Cluster Validation

A cluster summary and histogram of average values across the selected metrics (revolving balance, utilization ratio, and transaction count) showed clear and meaningful differences between clusters and aligned with our expected customer profiles. These patterns supported a four-cluster approach (K = 4), a choice supported by elbow analysis.

[1] Ilmad Dabbura, Towards Data Science , 2017

We used cluster-level summaries for both scaled and unscaled data versions to illustrate relative differences in behavior by cluster, as well as absolute averages. Their clusters map to the following customer profiles:

K-Means Cluster	Customer Profile	Defining Characteristics
0	Stressed Borrowers/Revolvers	High utilization ratio; high revolving balances
1	Everyday User	Moderate utilization ratio; moderate transaction activity
2	Intermediate/Active Revolvers	High revolving balances; lower utilization than Stressed Borrowers
3	Pay-in-Full Customers/Gamers	Low/zero revolving balances; frequent transactions

Scaled Cluster Histogram

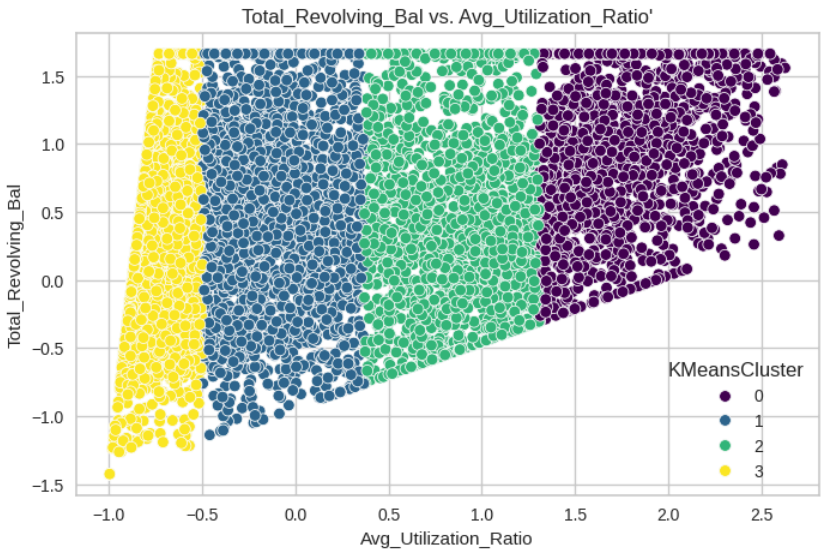
KMeansCluster	count	utilization_mean	transcnt_mean	revolving_mean
0	1529	1.757510	-0.063836	0.916669
1	2200	-0.122921	0.008751	0.429567
2	1821	0.864948	0.079251	0.401374
3	4577	-0.872160	-0.014412	-0.672392

UnScaled Cluster Histogram

KMeansCluster	count	utilization_mean	transcnt_mean	revolving_mean
0	1529	0.759400	63.360366	1909.850883
1	2200	0.241007	65.064091	1512.888636
2	1821	0.513340	66.718836	1489.912685
3	4577	0.034458	64.520428	614.850339

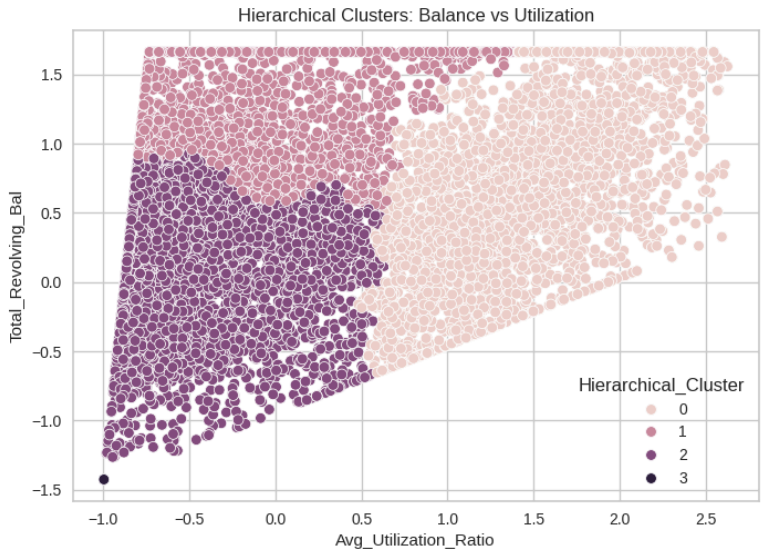
Finally, we confirmed our framework by mapping our clusters on a 2-D scatter plot comparing revolving balance and average utilization ratio. This illustration further supported the cluster decision, as it has clearly separated cluster groups, almost no overlap across groups, and not too much striping, which may suggest one dominating variable. These patterns align with expected differences in credit usage and financial stress and remain consistent whether the data are scaled or unscaled.

While we initially expected a fully dormant customer group might emerge, the results instead revealed meaningful variation among customers with high revolving balances, distinguishing more and less financially constrained borrowers based on utilization ratio. Overall, this suggests our differences across customers are more explained by credit stress than disengagement/attrition.



Hiercharchal Validation

To further validate the results from the K-Means Analysis, we repeated the segmentation using hierarchical clustering focusing on revolving balance and average utilization ratio.



While hierarchical clustering identified a more distinct low-engagement (dormant) customer group, it reinforced the same core insight as K-Means: meaningful differences among revolving customers are driven by relative financial strain, with utilization ratio distinguishing more financially constrained borrowers from less constrained users who tend to carry balances but repay more consistently.

DBSCAN

To challenge the assumptions of the K-Means model, we implemented **HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise). Unlike K-Means, which assumes spherical clusters of similar size, HDBSCAN searches for "islands" of high density. We utilized a slightly variant three-variable feature to demonstrate similar outcomes with variant metrics (Total_Revolving_Bal, Total_Trans_Ct, and Credit_Limit). Our testing followed a three-tier parameter strategy:

```
PARAMETER COMPARISON
    Name  min_cluster_size  min_samples  Clusters  Noise_Count Noise_Pct
  Strict               200           30         2         1255     12.4%
Moderate               100           10         4         2810     27.7%
 Relaxed                50            5         5         2186     21.6%
```

Results and Cluster Characteristics:

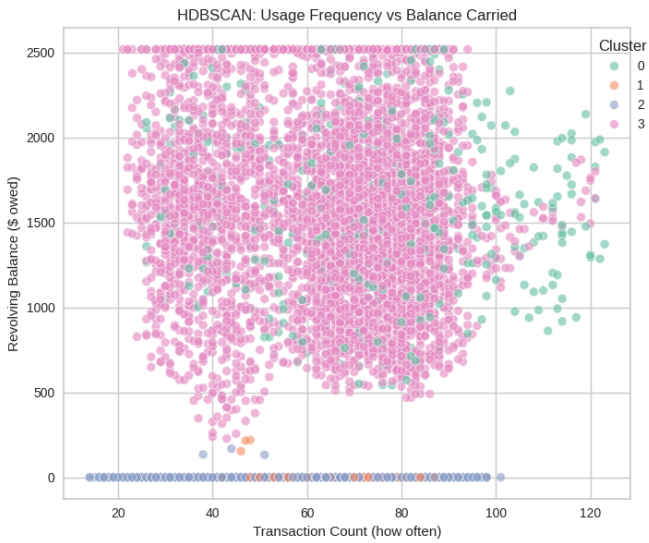
The HDBSCAN analysis identified four clusters with significant heterogeneity in size:

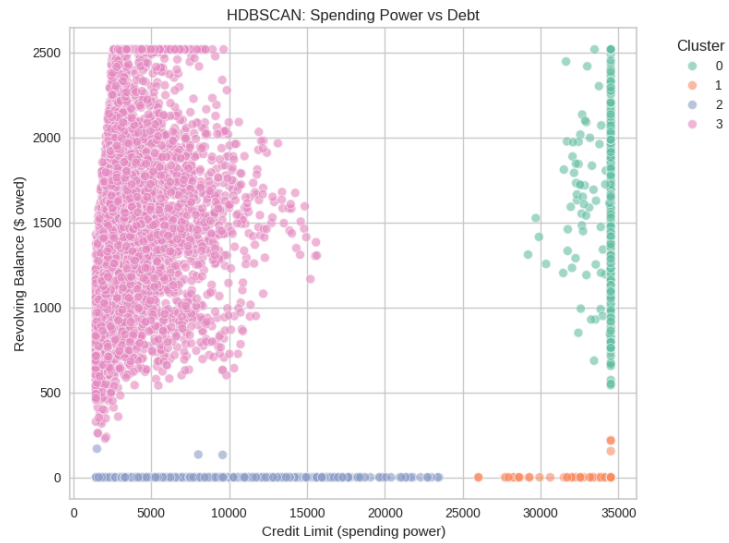
- **Cluster 0 (Pink - 80.4%):** The dominant "general population" cluster. This group forms a massive, dense cloud across almost all transaction counts and balance levels.
- **Cluster 1 (Blue - 6.4%):** "Low-balance transactors" who pay in full monthly. These are clearly visible along the y=0 axis (zero revolving balance).
- **Cluster 2 (Gray - 7.2%):** "Moderate users" with mid-range credit limits.
- **Cluster 3 (Green - 6.0%):** "High Credit Limit" customers. These appear as a distinct vertical band on the far right of the "Spending Power vs Debt" chart, representing the $30,000+ credit limit cap.

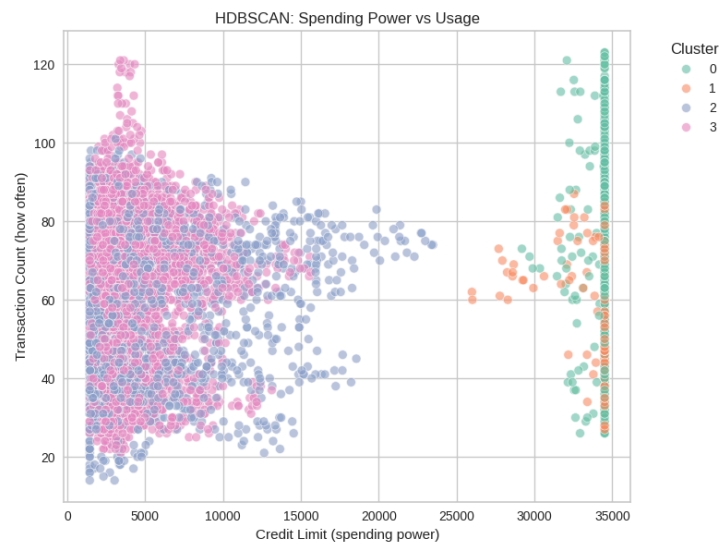
The "Clean Clusters" Analysis (Noise Removal)

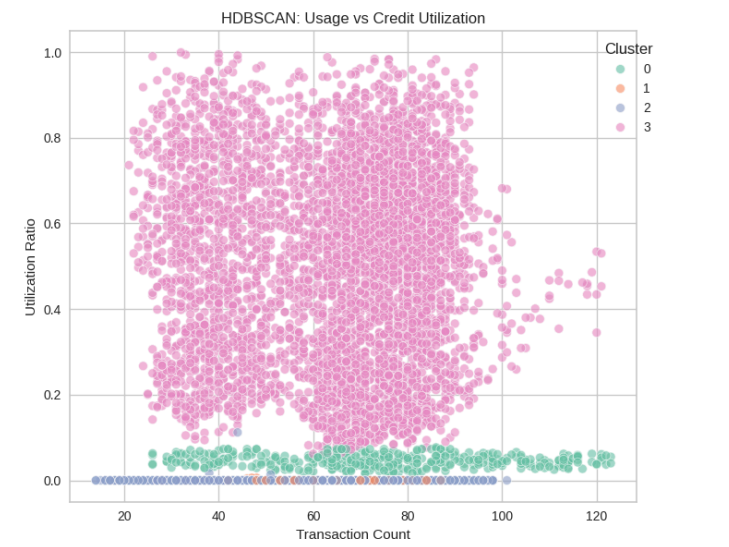
A critical step in our analysis was visualizing the data with noise points removed to see if a clearer structure emerged.

- **Persistence of Imbalance:** Even when excluding noise, the dominant cluster (Pink) remains a massive, heterogeneous block rather than a tightly defined segment. This confirms that the imbalance is inherent to the data's density, not caused by outliers.

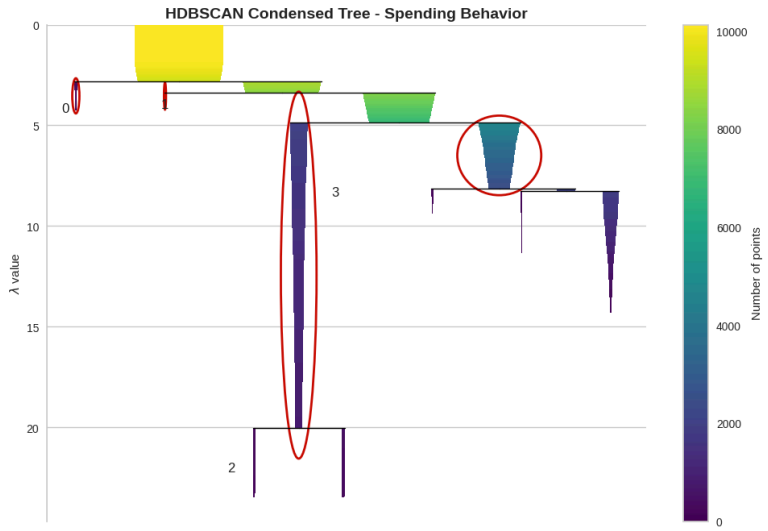








- **Algorithmic Structural Challenges:** The **HDBSCAN Condensed Tree** visualization reveals the root of the challenge: one massive cluster (represented by the dominant yellow bar at the top of the hierarchy) contains over 8,000 points. Smaller clusters (labeled 0, 1, 2, and 3) only emerge as distinct branches at much higher density thresholds (**λ** values).The majority of customers exist in a single "density cloud" that cannot be easily partitioned using this method.



Technical Challenges & Sensitivity

Our implementation highlighted three significant hurdles:

1. **Extreme Sensitivity:** Reducing (min_cluster_size) from 100 to 50 shattered the segments into 15+ micro-clusters, making results too granular for a marketing team to use.
2. **The Gradient Problem:** Our charts (especially **Usage vs Credit Utilization**) show that customers exist on a continuous spectrum. HDBSCAN expects "valleys" of empty space between clusters; because our data is a continuous "cloud," the algorithm forces an artificial separation or creates one catch-all group.
3. **Marketing Actionability:** A segment containing 80% of customers lacks the specificity required for targeted campaigns.

Conclusion: Why We Selected K-Means

While HDBSCAN successfully isolated edge cases like high-limit customers and full-payers, it failed to provide a meaningful breakdown of the bulk of the customer base. These findings validate our final selection of **K-Means**, which partitions the continuous data into more balanced and operationally useful segments for strategic marketing.

- For K-Means, clustering into four groups (as determined through the elbow method), proved most appropriate both statistically and through generated visualizations
- The largest customer distinction came from financial/credit stress, as clusters differentiated “high stress” borrowers from “lower stress” customers that utilized less credit, albeit both having revolving balances. Credit strain was a continuous gradient rather than a divisively separated distribution.

- DBSCAN revealed that most customers exist within the same “density cloud” in regards to behavior, making this clustering method overall less effective in generating usable insights. Although it did reveal outliers (like high credit limits and customers that pay in full), most of the data shaded pink was in a single cluster

Recommendations

- **Differentiate customers with revolving balances by credit strain:**
    - Across methods, utilization ratio consistently distinguishes more financially constrained borrowers from less constrained ones. The results suggest the bank should direct hardship programs or payment plans toward higher-strain borrowers, while focusing managed credit line increases on customers with lower utilization.
- **Prioritize rewards and premium products for Pay-in-Full and Everyday Users:**
    - Because these segments generate less value through carried balances, profit strategies should focus on cross-selling, premium customer experiences that support annual fees, and retained loyalty through rewards.
- **Avoid Overtargeting in Future Customer Analysis and Marketing Campaigns**
    - DBSCAN suggests customers exist on a continuum of behavioral metrics. K-Means-based segmentation focused on patterns of behavior should continue to guide future customer analysis.
- **Create a Credit Utilization-Based Trigger for Company Intervention**
    - Because clustering results show that customer behavior changes gradually rather than being distinctly segmented, early intervention should include rising or shrinking a customer’s credit utilization that prevents customers from “digging a deeper hole” before overdrafting or delinquency.